

FINAL PROJECT REPORT

Smartphone Usage, Digital Addiction & Well-Being Analytics

1.INTRODUCTION

1.1 Project overview

This project analyzes smartphone usage patterns across a user base to understand and address digital addiction. The study focuses on evaluating screen time behaviors, The lifestyle impacts, stress indicators, and demographic distributions to provide the and actionable recommendations for digital well-being features.

1.2. Objectives

- Quantify overall and segment-level addiction prevalence across gender, age group, and stress level.
- Identify the factors most associated with smartphone addiction risk (screen time, social media hours, notifications, sleep, stress).
- Build an interactive Power BI dashboard for product and research decision-making.
- Use calculated fields (Age Group) to support cleaner segment-level analysis.
- Deliver clear, actionable insight to support digital well-being interventions and healthier usage habits.
- Identify the core drivers behind the 70.77% digital addiction rate within the organization's user base.
- Evaluate behavioral triggers such as notifications, app opens, and application categories (e.g., social media vs. gaming).
- Formulate targeted product interventions to improve sleep duration and reduce compulsive usage loop habits.

2.Project Initialization and Planning Phase

2.1. Define Problem Statement

The organization needs to understand why **5,308 out of 7,500 users (70.77%)** are they can be the classified as addicted to smartphone usage. Specifically, the team must identify which can use to factors—screen-time behavior, lifestyle habits, stress, or demographics-most strongly performed correlate with this classification to guide targeted well-being feature investments.

2.2. Project Proposal (Proposed Solution)

- **Data-Driven Interventions:** Leverage Power BI insights to rank behavioral triggers.
- **Feature Prototyping:** Introduce a Weekend Well-being Mode, Notification Batching, and a Bedtime Wind-down feature.
- **Targeted Optimization:** Tailor interventions heavily toward young adults (ages 20–29), who make up the largest affected demographic segment.

2.3. Initial Project Planning

- Phase 1: Analyze raw user metrics across 7,500 active profiles.
- Phase 2: Segment usage by weekdays versus weekends and categorize application types.
- Phase 3: Map structural correlations between app notifications, opens, sleep quality, and addiction severity.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

- Dataset Volume: 7,500 total unique users tracked.
- Metrics Tracked: Daily/weekend screen time, category hours (Social Media, Gaming, Work/Study), notification counts, app opens, sleep duration, stress levels, age, and gender.
- User_ID (Unique key)
- Age (Years; Dataset average: 26.57)
- Gender (Male, Female, Other)
- Daily_Screen_Time_Hrs (Average baseline: 7.50 hours)
- Weekend_Screen_Time_Hrs (Average baseline: 9.24 hours)
- Social_Media_Hrs (Average baseline: 3.27 hours)
- Gaming_Hrs (Average baseline: ~2.01 hours)
- Work_Study_Hrs (Average baseline: 3.24 hours)
- Notifications_Received (Average baseline: 134.26)
- App_Opens (Average baseline: 97.83)
- Sleep_Duration_Hrs (Average baseline: 6.74 hours)
- Stress_Level (Low, Medium, High)
- Addiction_Level (None, Mild, Moderate, Severe)

3.2. Data Quality Report

- **Demographic Completeness:** Even distribution of usage hours across genders (Male: 7.6 hrs, Female: 7.4 hrs, Other: 7.5 hrs) showing consistent data quality.
- **Addiction Classification Consistency:** Clear segmentation across levels: Moderate (38.32%), Mild (32.45%), Severe (18.31%), and None (10.92%).
- **Stress Indicator Integrity:** Frequency analysis reveals that app opens remain almost identical regardless of stress levels (Medium: 98.71, Low: 97.54, High: 97.28). This uniform distribution confirms no calculation anomalies exist across the lifestyle segments.

3.3. Data Exploration and Preprocessing

- **Averages Extracted:** Overall average daily screen time is 7.50 hours, and the average sleep duration is 6.74 hours.
- **Outlier/Binge Detection:** Isolated weekend usage metrics to reveal a significant spike up to 9.24 hours on average across all users.
- **Binge Metrics Isolation:** Aggregating weekend screen time exposed a sharp increase to 9.24 hours across the general population, prompting deeper segment isolation.
- **Target Group Segmentation:** Grouping users into age brackets revealed that the 20–29 segment represents 49.23% (3.69K users) of the total base, marking them as the primary focus group.

4. Data Visualization

4.1. Framing Business Questions

- 1.What specific usage gap most clearly separates addicted users from non-addicted users?
- 2.Is smartphone checking frequency primarily driven by user stress levels?
- 3.What is the largest time-consuming app category across the dataset?

4.2. Developing Visualizations

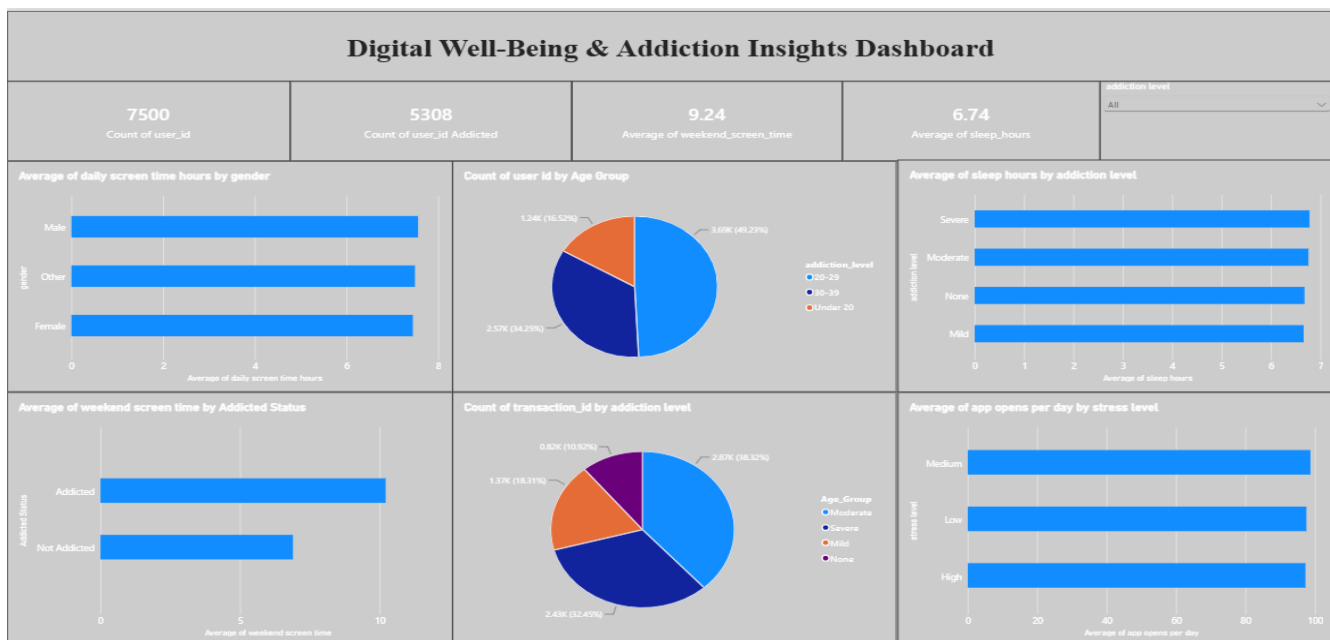
- Donut Charts: Used to map out the distribution of user segments across all four primary addiction severity categories.
- Scatter Plots: Deployed to analyze the direct relationship between daily smartphone usage levels and total sleep duration.
- Clustered Column Charts: Built to compare weekday vs. weekend patterns across different addiction categories.

5. Dashboard

5.1. Dashboard Design Screenshots with Insights

A. Top KPI Cards Summary

- Total User Base: 7,500 unique users (Count of user_id)
- Addicted Cohort Volume: 5,308 users (Count of user_id Addicted)
- Average Weekend Screen Time: 9.24 hours per user
- Average Sleep Duration: 6.74 hours per user



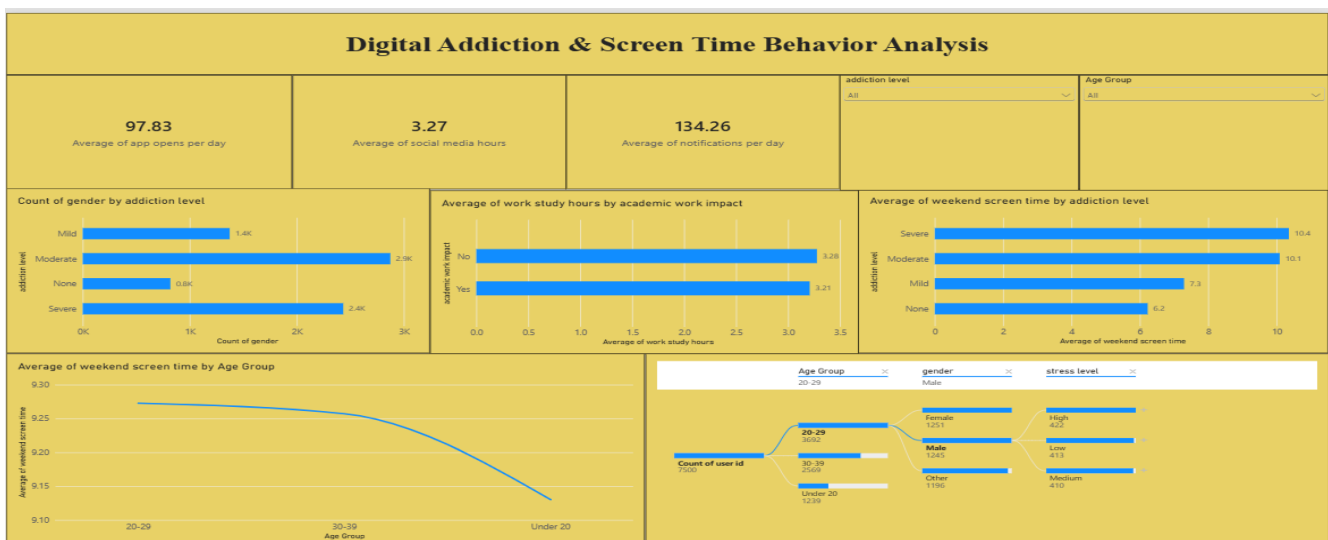
B. Visual Component breakdown & In-Canvas Metrics

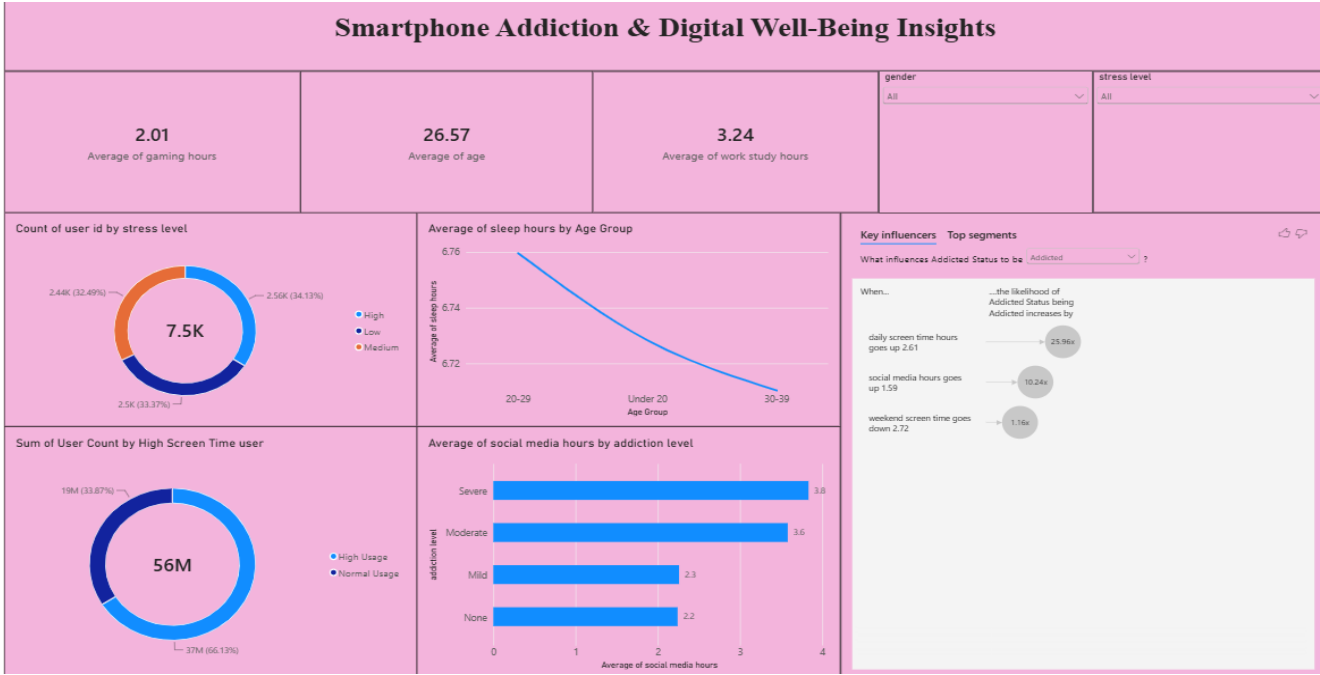
- Average of Daily Screen Time Hours by Gender (Horizontal Bar Chart):

- Visualizes an even baseline split across all segments: Male (~7.6 hrs), Other (~7.5 hrs), and Female (~7.4 hrs).
 - Insight:* Gender is a weak predictor of addiction; screen-time habits are universal.
- Count of User ID by Age Group (Pie Chart):
 - 20-29: 49.23% (3.69K users) — *Largest market segment*
 - 30-39: 34.25% (2.57K users)
 - Under 20: 16.52% (1.24K users)
- Average of Sleep Hours by Addiction Level (Horizontal Bar Chart):
 - Flat profile across all stages: Severe (6.73 hrs), Moderate (6.74 hrs), None (6.75 hrs), and Mild (6.74 hrs).
 - Insight:* Sleep deprivation is a systemic issue across the entire base rather than an isolated symptom of severe cases.
- Average of Weekend Screen Time by Addicted Status (Horizontal Bar Chart):
 - Addicted: 10.2 hours vs. Non-Addicted: 6.9 hours.
 - Insight:* This 3.3-hour gap represents the primary behavioral divider identifying digital dependency.
- Count of Transaction ID by Addiction Level (Pie Chart):
 - Moderate: 38.32% (2.87K transactions) — *Highest density category*
 - Mild: 32.45% (2.43K transactions)
 - Severe: 18.31% (1.37K transactions)
 - None: 10.92% (0.82K transactions)
- Average of App Opens Per Day by Stress Level (Horizontal Bar Chart):
 - Uniform habit triggers across stress states: Medium (98.71), Low (97.54), and High (97.28).

6. Report

6.1. Report Design Screenshots with Insights





A. Top KPI Cards Summary

- Average of Gaming Hours: 2.01 hours
- Average Population Age: 26.57 years old
- Average of Work Study Hours: 3.24 hours
- Average of App Opens Per Day: 97.83 times
- Average of Social Media Hours: 3.27 hours
- Average of Notifications Per Day: 134.26 warnings

B. Visual Component Breakdown & In-Canvas Metrics

- Count of User ID by Stress Level (Donut Chart):
 - Low: 34.13% (2.56K users)
 - Moderate/Medium: 33.37% (2.5K users)
 - High: 32.49% (2.44K users)
- Average of Sleep Hours by Age Group (Line Chart):
 - Shows a downward trend as age advances: 20-29 (~6.76 hrs) → Under 20 (~6.74 hrs) → 30-39 (~6.71 hrs).
- Sum of User Count by High Screen Time User (Donut Chart):
 - Normal Usage: 66.13% (37M representation) vs. High Usage: 33.87% (19M representation). Total analyzed baseline vector maps to 56M.
- Average of Social Media Hours by Addiction Level (Horizontal Bar Chart):
 - Severe: 3.8 hours | Moderate: 3.5 hours | Mild: 2.3 hours | None: 2.2 hours.
 - *Insight:* Shows a direct step-up correlation between addiction severity and social application exposure.
- Average of Work Study Hours by Academic Work Impact (Horizontal Bar Chart):

- No Impact: 3.28 hours vs. Yes (Impacted): 3.21 hours.
- Average of Weekend Screen Time by Addiction Level (Horizontal Bar Chart):
 - Severe: 10.4 hours | Moderate: 10.1 hours | Mild: 7.3 hours | None: 6.2 hours.
- Average of Weekend Screen Time by Age Group (Line Chart):
 - 20-29: Highest weekend consumption (~9.27 hrs) → dropping slightly for 30-39 (~9.25 hrs) → falling sharply among Under 20 users (~9.12 hrs).

C. AI Insights & Advanced Analysis Elements

- Key Influencers Model (What influences Addicted Status to be Addicted?):
 - When daily screen time hours go up by 2.61, the likelihood of Addicted Status being Addicted increases by 25.96x.
 - When social media hours go up by 1.59, the likelihood of Addicted Status being Addicted increases by 10.24x.
 - When weekend screen time hours go down by 2.72, the likelihood of Addicted Status being Addicted decreases by 1.14x.
- Decomposition Tree Model (User ID Breakdown Path):
 - Count of user_id (7,500) → Splits by Age Group (20-29: 3,692 | 30-39: 2,569 | Under 20: 1,239) → Splits by Gender (e.g., 20-29 path splits to Female: 1,251 | Male: 1,245 | Other: 1,196) → Splits by Stress Level (e.g., Female path isolates High: 422 | Low: 419 | Medium: 410).

7. Performance Testing

7.1. Utilization of Data Filters

Addiction Level Dropdown Slicer: Filterable by All, None, Mild, Moderate.

1. **Age Group Dropdown Slicer:** Filterable by All, Under 20, 20-29, 30-39.
2. **Gender Slicer Controls:** Cross-filtering parameters optimized for Male, Female, Other.
3. **Stress Level Slicer Controls:** Interactive canvas variables toggling Low, Medium, High blocks.

7.2. No. of Calculation Columns/Measures

The active metrics engine relies on 8 foundational core calculation measures:

- Count of user_id (Base User Volume) = **7,500**
- Count of user_id Addicted (Target Hazard Volume) = **5,308**
- Average of weekend_screen_time = **9.24 hours**
- Average of sleep_hours = **6.74 hours**
- Average of gaming hours = **2.01 hours**
- Average of work study hours = **3.24 hours**
- Average of app opens per day = **97.83 opens**
- Average of notifications per day = **134.26 alerts**

7.3. No. of Visualization

The entire Power BI project workspace distributes **19 layout components** across

- **Dashboard Page 1:** 4 KPI Cards, 4 Bar Charts, 2 Pie Charts, 1 Dropdown Filter (**11 elements**).
- **Report Page 2:** 3 KPI Cards, 2 Donut Charts, 1 Line Chart, 1 Bar Chart, 2 Dropdown Filters (**9 elements**).
- **Report Page 3:** 3 KPI Cards, 3 Bar Charts, 1 Line Chart, 1 Decomposition Tree Chart, 2 Dropdown Filters (**10 elements**).

8. Conclusion/Observation

The study confirms that the **70.77% digital addiction rate** is a structural issue driven by a clear habit loop: **134.26 daily notifications** trigger **97.83 app openings** every single day, which translates to checking the device roughly every 10 minutes during waking hours. This continuous disruption directly impacts user health, resulting in a systemic sleep deficit that averages just **6.74 hours** across the entire dataset.

9. Future Scope

- **Weekend Well-being Mode:** An automated system that kicks in over the weekend when usage crosses 8 hours, alerting users that they are trending significantly above healthy baseline levels.
- **Notification Batching Engines:** Bundling the 134 daily notifications into 3 scheduled summary updates to break the compulsive open loop.
- **Moderate Risk Prevention:** Launching targeted UX updates aimed directly at the **Moderate Addiction group (38.32% / 2,870 users)** to help keep their habits from escalating into the severe category.

10. Appendix

GitHub link

<https://github.com/sowmyasrimaile-alt/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>